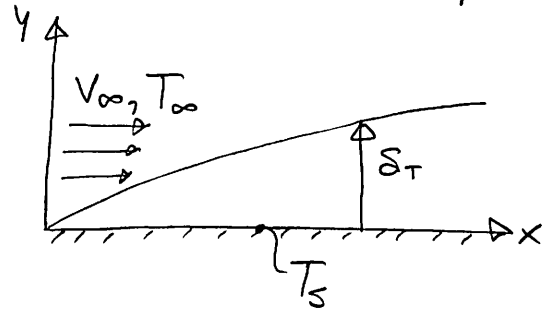


Heat Transfer

Back from our interesting hydrodynamic excursion, we are interested in the convective heat transfer from the flat plate:

$$h = \frac{q''|_{y=0}}{\Delta T} = \frac{q''|_{y=0}}{T_s - T_\infty} = ?$$



Note, here we have the thermal boundary layer thickness, δ_T .

$\delta \equiv$ thickness where the velocity gradient exists

$\Leftrightarrow \delta_T \equiv$ thickness where the temperature gradient exists.

To solve for this second boundary layer, we need to solve the energy equation for the fluid flow.

$$\underbrace{u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y}}_{\text{Convection of energy}} = \alpha \underbrace{\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)}_{\text{Conduction of energy}}$$

$\Rightarrow$ Note, we always had $\nabla^2 T = 0$ in conduction (R.H.S.)

For the $T_s = \text{constant}$ case, $\frac{\partial^2 T}{\partial x^2} \ll \frac{\partial^2 T}{\partial y^2}$

Boundary conditions: 1) $T(y=0) = T_s$
2) $T(y \rightarrow \infty) = T_\infty$

Non-dimensionalizing: Let $\theta = \frac{T - T_s}{T_\infty - T_s}$

$$u \frac{\partial \theta}{\partial x} + v \frac{\partial \theta}{\partial y} = \alpha \frac{\partial^2 \theta}{\partial y^2}$$

Note the similarity of the boundary layer equations

Hydrodynamic (δ)	Thermal (δ_T)
$u \frac{\partial \bar{u}}{\partial x} + v \frac{\partial \bar{u}}{\partial y} = \nu \frac{\partial^2 \bar{u}}{\partial y^2}$	$u \frac{\partial \theta}{\partial x} + v \frac{\partial \theta}{\partial y} = \alpha \frac{\partial^2 \theta}{\partial y^2}$
B.C.'s: $\bar{u}(y=0) = 0$	$\theta(y=0) = 0$
$\bar{u}(y \rightarrow \infty) = 1$	$\theta(y \rightarrow \infty) = 1$
$\left. \frac{\partial \bar{u}}{\partial y} \right _{y \rightarrow \infty} = 0$	$\left. \frac{\partial \theta}{\partial y} \right _{y \rightarrow \infty} = 0$

So we can find an analogy to our fluids solution

We can first define a usefull property called the Prandtl number

$$Pr = \frac{\nu}{\alpha} \equiv \frac{\text{viscous (momentum) diffusion rate}}{\text{thermal (heat) diffusion rate}}$$

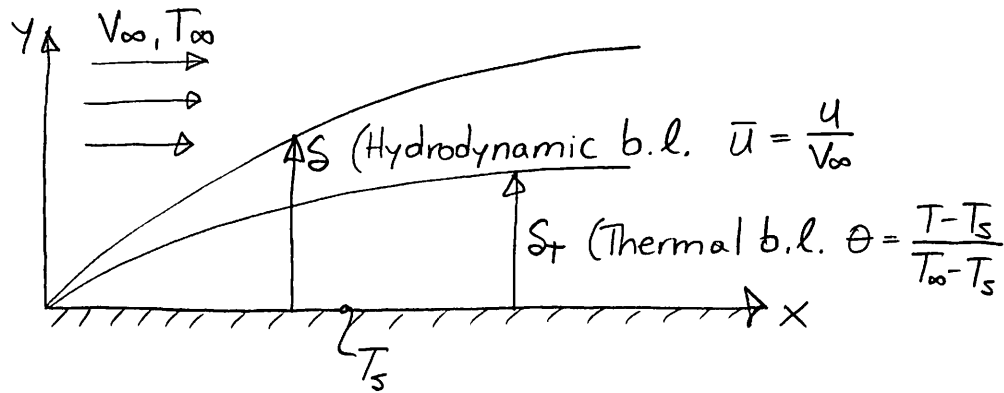
If $Pr = 1$, $\boxed{\theta = \bar{u}, \delta = \delta_T} \Rightarrow$ Hydrodynamic & thermal boundary layers overlap.

However, in real life, most fluids aren't $Pr = 1$.

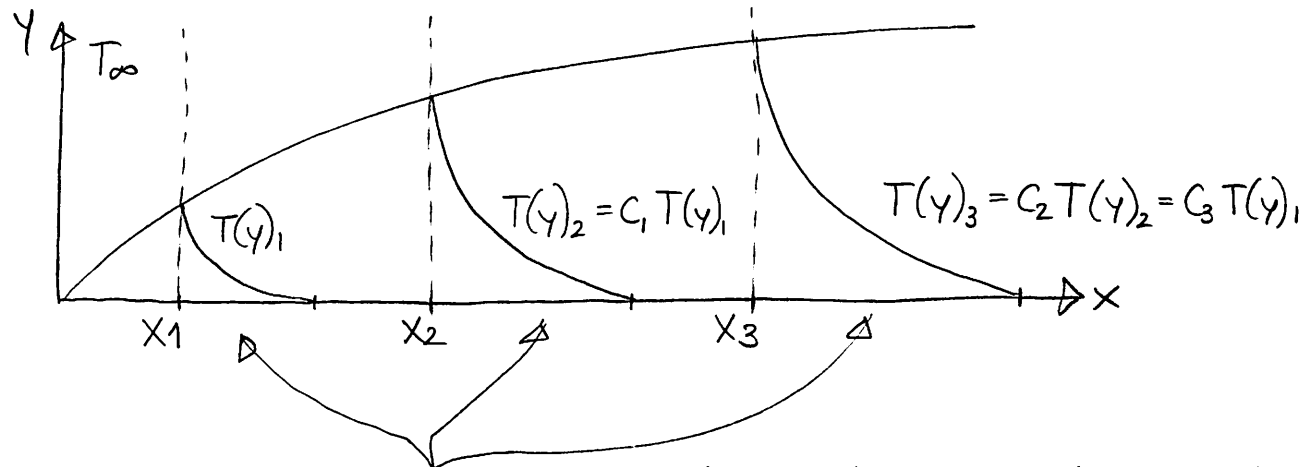
$Pr \approx 0.001$ for liquid metals ≈ 0.015 for mercury $\approx 0.7 - 0.8$ for air $\approx 4 - 5$ for refrigerants ≈ 7 for water $\approx 100 - 40,000$ for engine oil $\approx 1 \times 10^{25}$ for the earth's mantle	$\left. \begin{array}{l} \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \end{array} \right\} \delta < \delta_T$ $\left. \begin{array}{l} \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \end{array} \right\} \delta > \delta_T$
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If $U \neq \infty$:

Note, the case I drew here $\Leftarrow$ is for $Pr > 1$.



OK, so let's solve the thermal b.l. equation for $\alpha \neq \nu$
We will use the same similarity approach:



All three profiles look self-similar in nature.

Assume: $\Theta = f(\eta)$, $\eta = y \sqrt{\frac{V_\infty}{\nu x}}$ (same similarity variable as in the hydrodynamic case)

$$\frac{\partial \Theta}{\partial x} = \frac{\partial \Theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial x}$$

$$\frac{\partial \Theta}{\partial y} = \frac{\partial \Theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial y}$$

$$\frac{\partial^2 \Theta}{\partial y^2} = \frac{\partial^2 \Theta}{\partial \eta^2} \cdot \left(\frac{\partial \eta}{\partial y} \right)^2$$

Back substitute these into our energy equation PDE:

$$u \frac{\partial \Theta}{\partial x} + v \frac{\partial \Theta}{\partial y} = \alpha \frac{\partial^2 \Theta}{\partial y^2} \Rightarrow u \frac{\partial \Theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} + v \frac{\partial \Theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} = \alpha \frac{\partial^2 \Theta}{\partial \eta^2} \cdot \left(\frac{\partial \eta}{\partial y} \right)^2$$

Note, we also know that: $F = \int_0^\eta \phi d\eta$, $\bar{u} = \phi(\eta) = \frac{u}{V_\infty} = F'$

Back substituting and doing some algebra, we obtain:

$$\frac{\partial^2 \theta}{\partial \eta^2} + \frac{1}{2} F \cdot Pr \frac{\partial \theta}{\partial \eta} = 0$$

Using a trick: $Pr^{2/3} \cdot F(\eta) = F(\eta^*)$; $\eta^* = \eta Pr^{1/3}$
 $\frac{d\eta^*}{d\eta} = Pr^{1/3}$

Now our equation becomes:
 $\underbrace{F \text{ at } \eta^*, \text{ not } F \text{ times } \eta^*!}$

$$\frac{\partial^2 \theta}{\partial \eta^2} + \frac{1}{2} F(\eta^*) Pr^{1/3} \frac{\partial \theta}{\partial \eta} = 0$$

$$\frac{\partial^2 \theta}{\partial \eta^2} + \frac{1}{2} F(\eta^*) Pr^{2/3} \cdot \underbrace{\frac{\partial \theta}{\partial \eta Pr^{1/3}}}_{\frac{\partial \theta}{\partial \eta^*}} = 0$$

Multiplying through by $Pr^{-2/3}$

$$\underbrace{\frac{1}{Pr^{2/3}} \left(\frac{\partial^2 \theta}{\partial \eta^2} + \frac{1}{2} F(\eta^*) Pr^{2/3} \cdot \frac{\partial \theta}{\partial \eta^*} \right)}_{\frac{\partial^2 \theta}{\partial \eta^{*2}}} = 0$$

↓
 Note, this works because:

$$F = \frac{a_2 \eta^2}{2} \dots (\text{H.O.T.})$$

If we only use the lead term:

$$F(\eta) = \frac{a_2 \eta^2}{2}$$

$$F(\eta^*) = \frac{a_2 Pr^{2/3} \eta^{*2}}{2}$$

$$F(\eta^*) = Pr^{2/3} F(\eta)$$

Now our PDE becomes identical to our hydrodynamic ODE

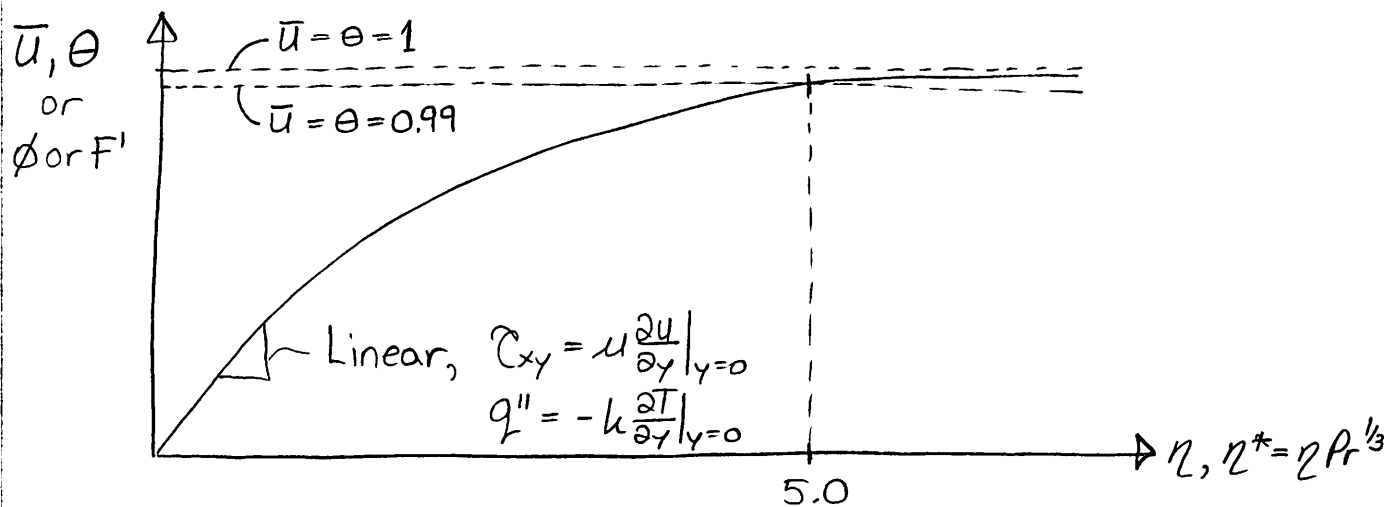
$$\boxed{\frac{\partial^2 \theta}{\partial \eta^{*2}} + \frac{1}{2} F(\eta^*) \frac{\partial \theta}{\partial \eta^*} = 0}$$

$$\Rightarrow \theta(\eta^*) \equiv \bar{u}(\eta)$$

Before, we had: $F''' + \frac{1}{2} FF'' = 0$

$$\left. \begin{array}{l} \text{B.C.'s : } 1) \theta(\eta^*=0) = 0 \\ \quad \quad 2) \theta(\eta^* \rightarrow \infty) = 1 \end{array} \right\} \theta = \frac{T - T_s}{T_\infty - T_s}$$

So we can now use the exact same Blasius solution plot that we had before.



$$\delta \sqrt{\frac{V_\infty}{\nu x}} = \eta(y = \delta) = 5.0 \quad (\text{Hydrodynamic b.l.})$$

$$Pr^{1/3} \delta_T \sqrt{\frac{V_\infty}{\nu x}} = \eta^*(y = \delta_T) = 5.0 \quad (\text{Thermal b.l.})$$

Taking the ratio of our two solutions above:

$$\boxed{\frac{\delta}{\delta_T} = Pr^{1/3}} \Rightarrow \text{Makes sense since the only difference is } \nu \text{ \& } \alpha \Rightarrow Pr = \frac{\nu}{\alpha}$$

Heat Transfer ($q'' = \text{heat flux}$)

$$\left. \begin{aligned} q''|_{y=0} = -k \frac{\partial T}{\partial y}|_{y=0} &\Rightarrow \Theta = \frac{T - T_s}{T_\infty - T_s}, \quad \partial \Theta = \frac{\partial T}{T_\infty - T_s} \\ \eta^* = \eta Pr^{1/3} &= y \sqrt{\frac{V_\infty}{\nu x}} \cdot Pr^{1/3} \\ \partial \eta^* &= \partial y \left(\frac{V_\infty}{\nu x} \right)^{1/2} Pr^{1/3} \end{aligned} \right\} \text{Back substitute}$$

$$\begin{aligned} q''|_{y=0} &= -k (T_\infty - T_s) \left(\frac{V_\infty}{\nu x} \right)^{1/2} Pr^{1/3} \frac{\partial \Theta}{\partial \eta^*} \bigg|_{\eta^*=0} \\ &= k \frac{(T_s - T_\infty)}{x} \underbrace{\left(\frac{V_\infty x}{\nu} \right)^{1/2}}_{Re_x^{1/2}} Pr^{1/3} \underbrace{F''(0)}_{a_2 = 0.332} \end{aligned}$$

$$\boxed{Nu_x = 0.332 Re_x^{1/2} Pr^{1/3}}$$

↳ Local Nusselt number for a flat plate in laminar flow.

$$q''|_{y=0} = \frac{k \Delta T}{x} Re_x^{1/2} Pr^{1/3} a_2, \quad Nu_x = \frac{hx}{k} = \frac{q''|_{y=0}}{\Delta T} \cdot \frac{x}{k} = a_2 Re^{1/2} Pr^{1/3} \quad (116)$$